Definition

- Markov-inverse-F measure (MiF)

$$D_{xy} = \sum_{i=1}^{\gamma} \frac{\alpha_i S_{x,y}^{(i)}}{H_\beta(P_x^{(i)}, P_y^{(i)})} = \sum_{i=1}^{\gamma} \frac{\alpha_i S_{x,y}^{(i)}(\beta P_x^{(i)} + (1-\beta)P_y^{(i)})}{P_x^{(i)} P_y^{(i)}}$$

$$= \sum_{i=1}^{\gamma} \frac{\alpha_i A_{x,y}^i (\beta \sum_{c=1}^{m} A_{x,c}^i + (1-\beta)\sum_{c=1}^{m} A_{y,c}^i)}{\sum_{c=1}^{m} A_{x,c}^i \sum_{c=1}^{m} A_{y,c}^i}$$

$$H_\beta(a,b) = \frac{ab}{\beta a + (1-\beta)b}$$

As weighted Harmonic Mean (F Measure), β: weight of F

A	Adjacency matrix
$A_{x,y}$	Element of A row: x, column: y
$S_{x,y}^{(i)} = A_{x,y}^i$	Number of the paths (routes) connecting the vertices x and y with i steps
$P_x^{(i)} = \sum_{c=1}^{m} A_{x,c}^i$	Number of the paths starting from the vertex x to any vertex with i steps

$$0 < \alpha_i < 1, \alpha_1 > \alpha_2 > \ldots, \sum_{i=1} \alpha_i = 1$$

$\alpha_1 + \alpha_2 + \ldots + \alpha_p$=(1/ coefficient[p]) + (1/ coefficient[p])^2 +... + (1/ coefficient[p])^p = 1

coefficient = {1, 1.618033988749895, 1.8392867552141607, 1.9275619754829254, 1.9659482366454855, 1.9835828434243263, 1.9919641966050354, 1.9960311797354144, 1.9980294702622872, 1.9990186327101014}

γ Small natural number to limit the step length

cf. Hiroyuki Akama, Maki Miyake, Jaeyoung Jung, Brian Murphy, 2015. Using Graph Components Derived from an Associative Concept Dictionary to Predict fMRI Neural Activation Patterns that Represent the Meaning of Nouns, PLoS ONE, doi: https://doi.org/10.1371/journal.pone.0125725